

Physical Chemistry (I)

Lecture 17

Special Lecture:

Protein Folding Theory



Phase Transitions

Ordered phase

energetically
favored

Interactions



Disordered phase

entropically
favored

$$S = k_B \ln W$$

Helmholtz and Gibbs Energies

Helmholtz energy $A = U - TS$

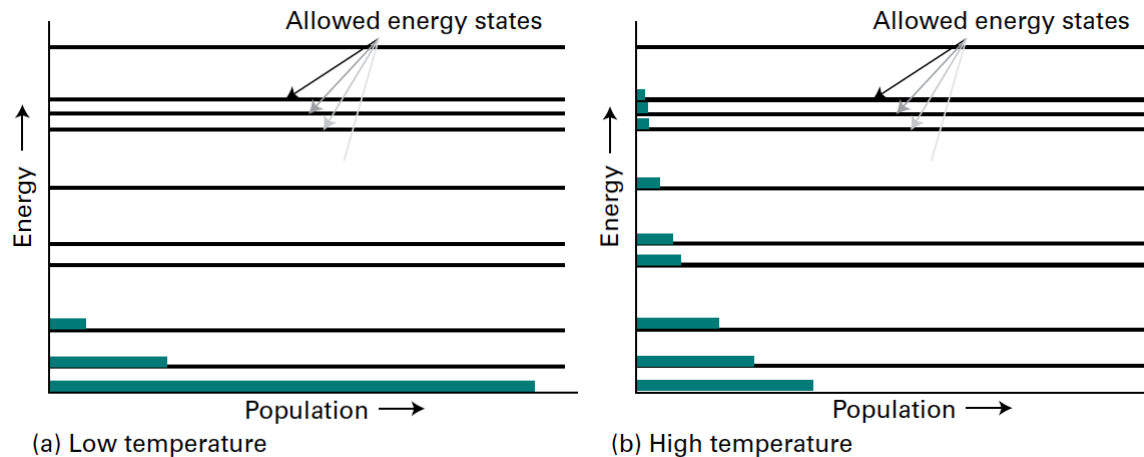
Gibbs energy $G = H - TS$

Both contributions taken into account

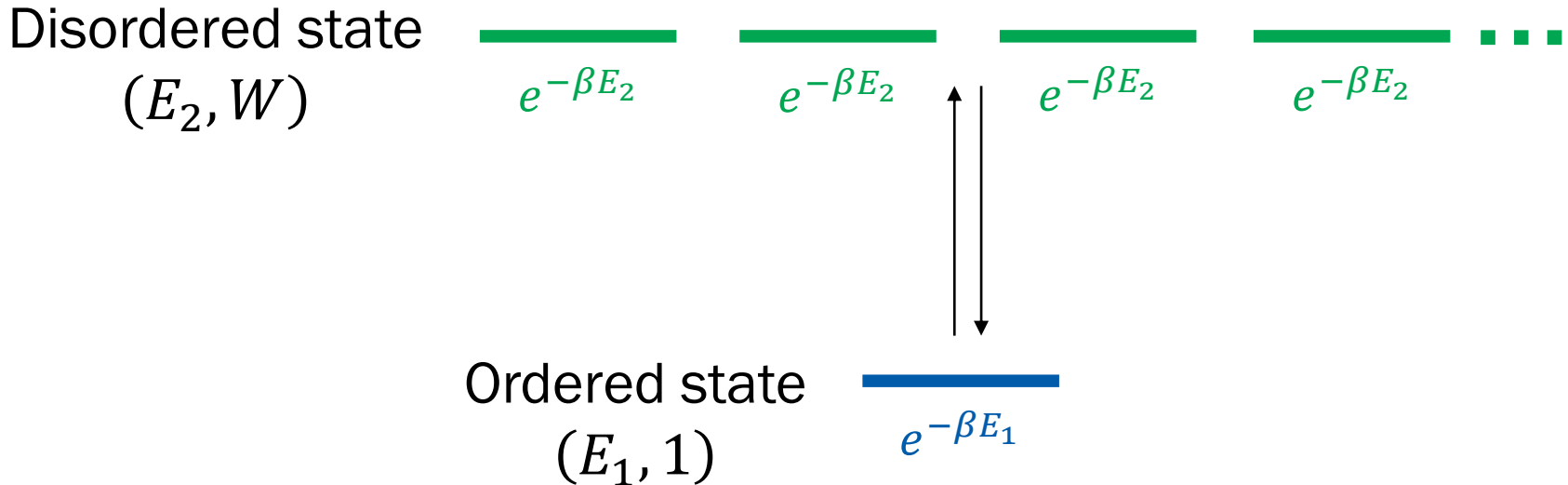
Boltzmann Distribution

$$\text{Probability} \propto e^{-E/k_{\text{B}}T} = e^{-\beta E}$$

(at thermal equilibrium)



Two-State Model

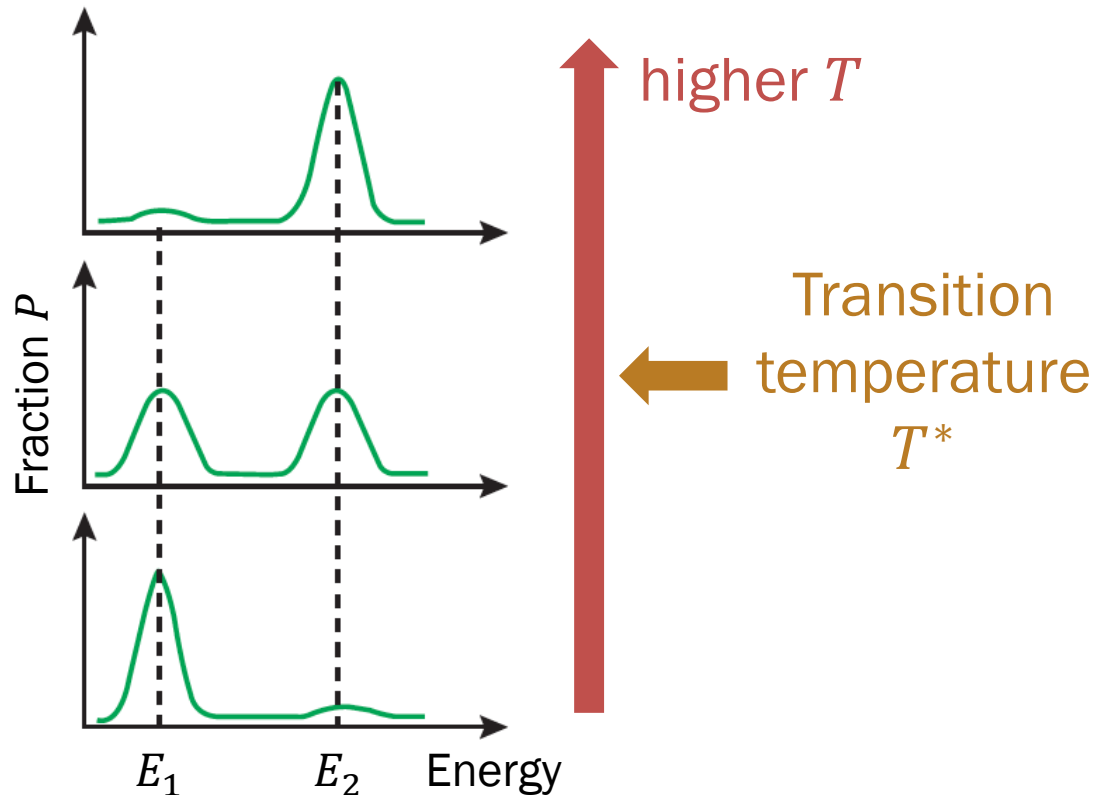


$$P_1 = \frac{e^{-\beta E_1}}{e^{-\beta E_1} + W e^{-\beta E_2}} = \frac{e^{-\beta A_1}}{e^{-\beta A_1} + e^{-\beta A_2}}$$

(First-Order) Phase Transition

$$P_1 = \frac{e^{-\beta A_1}}{e^{-\beta A_1} + e^{-\beta A_2}}$$

$$P_2 = \frac{e^{-\beta A_2}}{e^{-\beta A_1} + e^{-\beta A_2}}$$



Transition Temperature

$$P_1 = \frac{e^{-\beta A_1}}{e^{-\beta A_1} + e^{-\beta A_2}}, \quad P_2 = \frac{e^{-\beta A_2}}{e^{-\beta A_1} + e^{-\beta A_2}}$$

At the transition temperature, $P_1 = P_2$

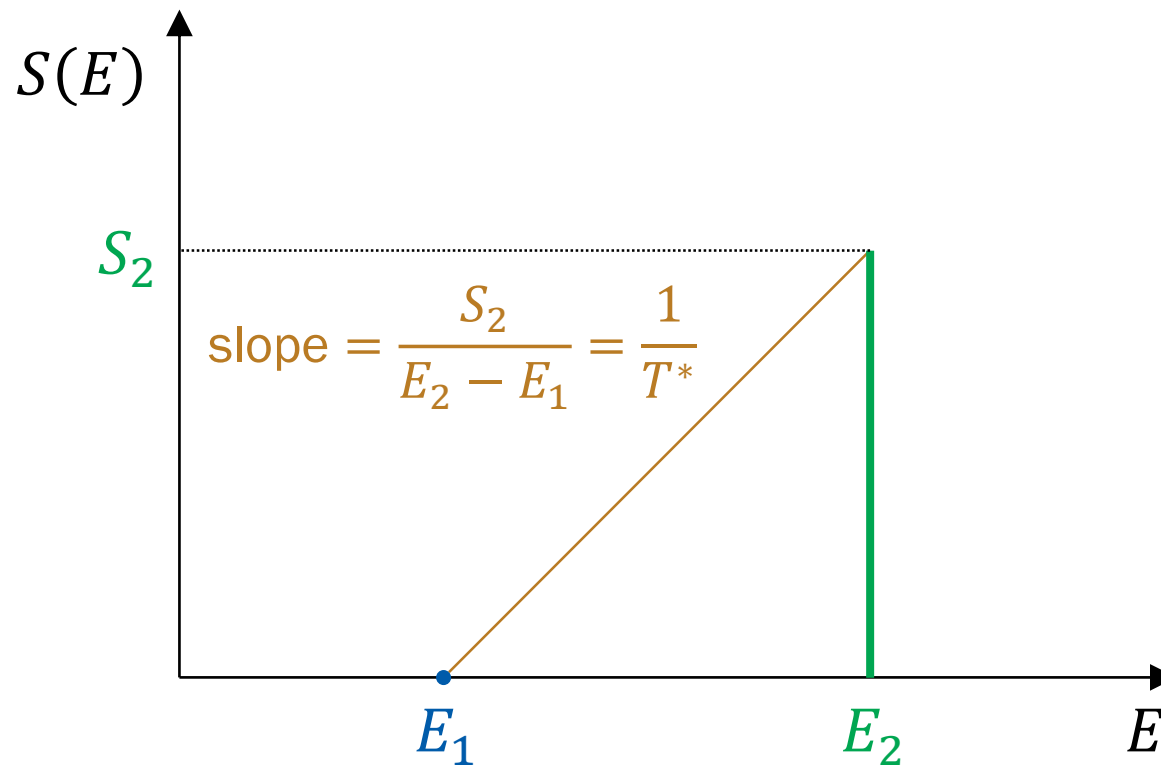
$$e^{-\beta A_1} = e^{-\beta A_2}$$

Hence,

$$A_1 = E_1 = E_2 - T^* S_2 = A_2$$

$$T^* = \frac{E_2 - E_1}{S_2} = \frac{E_2 - E_1}{k_B \ln W}$$

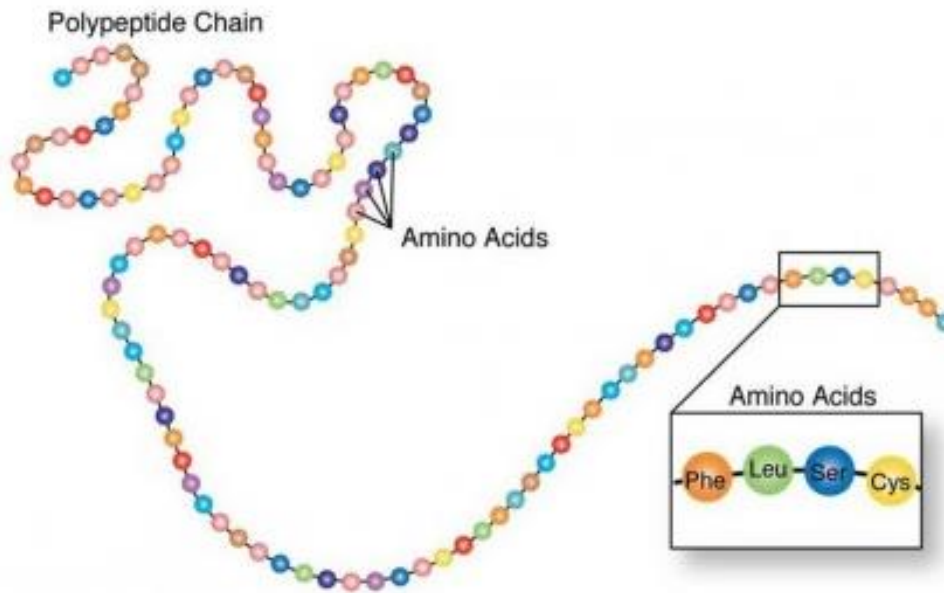
Another Derivation of T^*



Now Let's Talk About **Proteins** ...



Proteins are Polymer Chains



Proteins are Polymer Chains

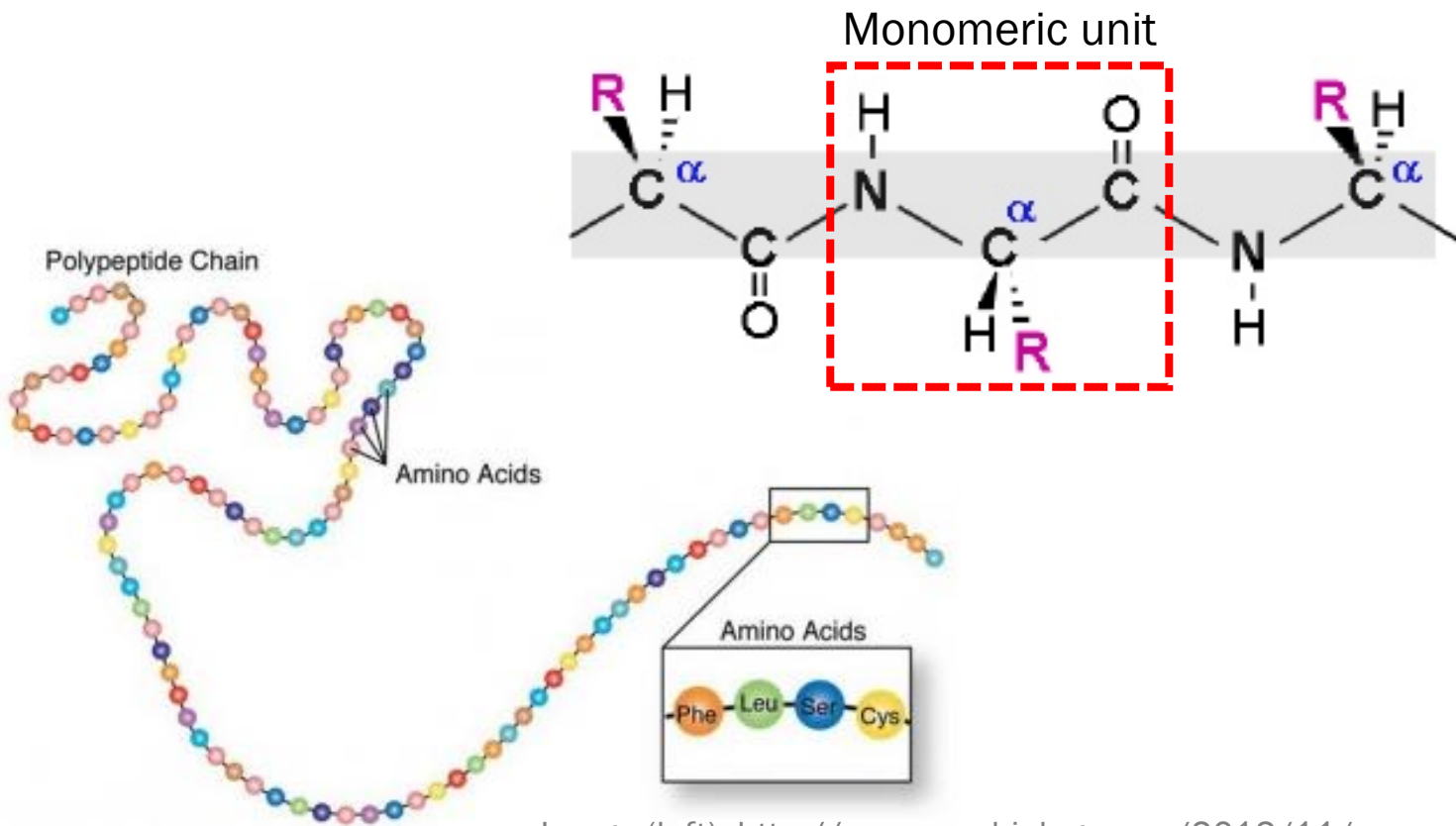


Image (left): <http://www.mcqbiology.com/2012/11/mcq-on-amino-acids.html>

Image (right): <https://www.slideshare.net/MUBOSScz/14-proteins>

Proteins are Polymer Chains

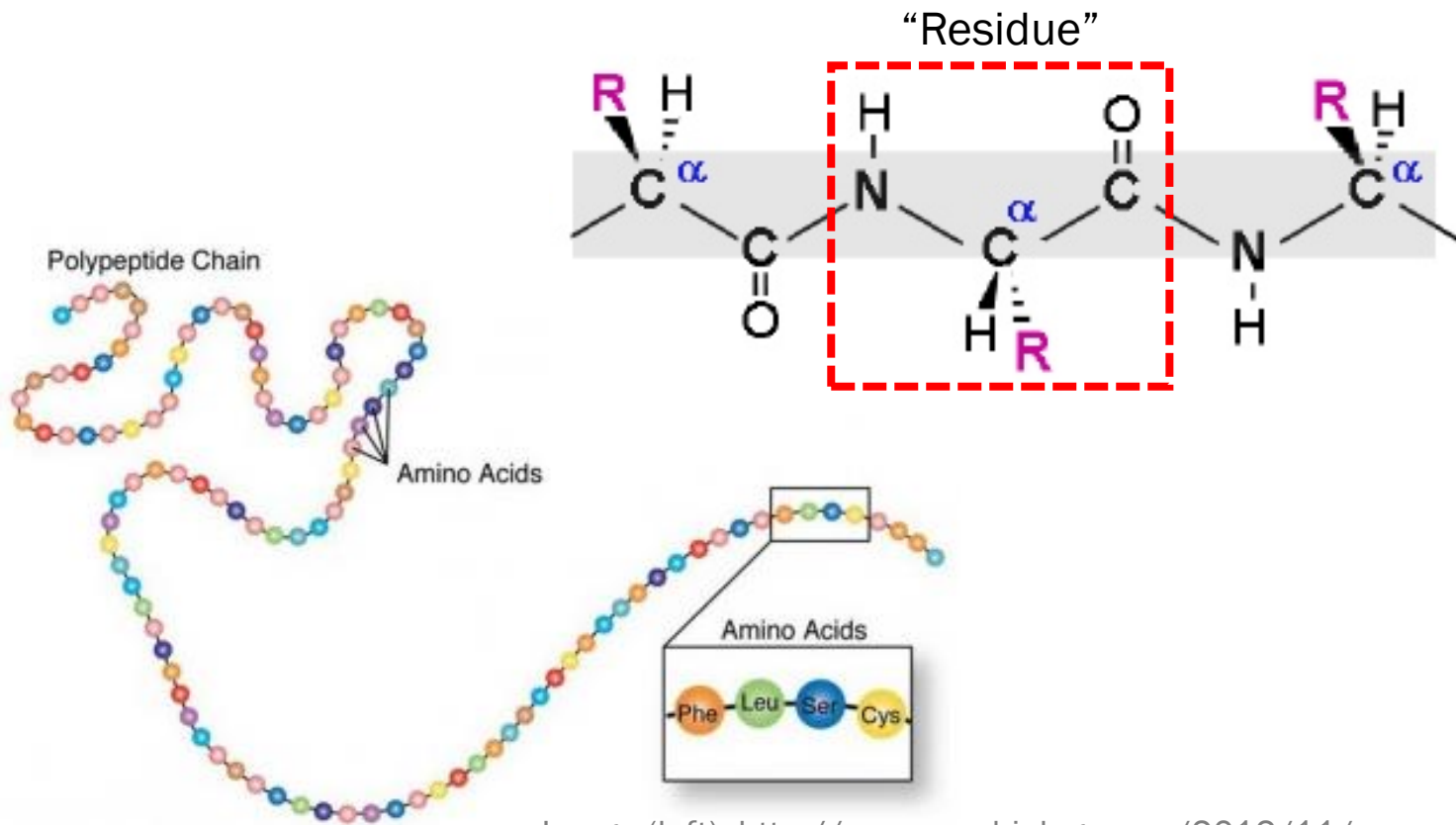


Image (left): <http://www.mcqbiology.com/2012/11/mcq-on-amino-acids.html>

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Proteins are Polymer Chains

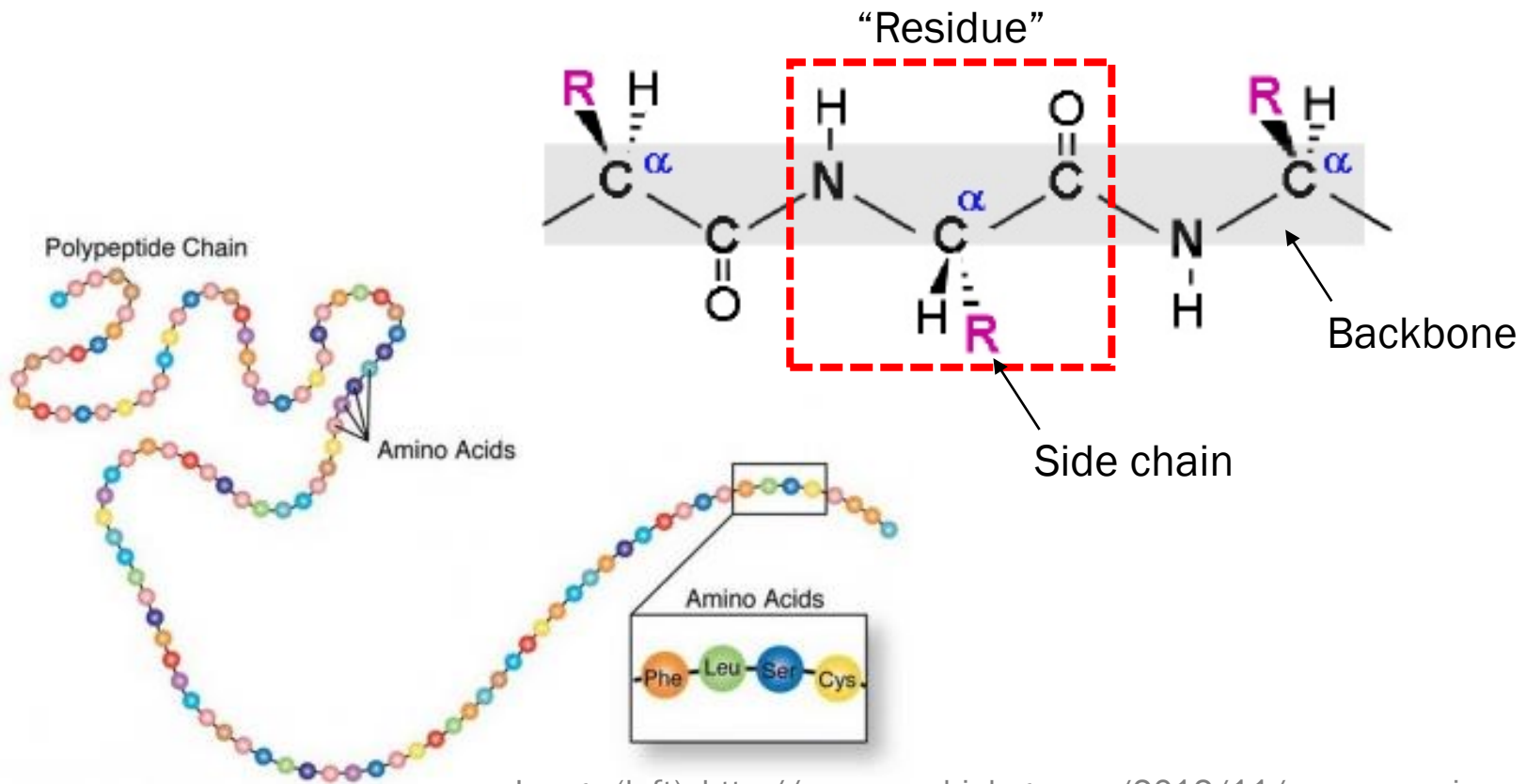
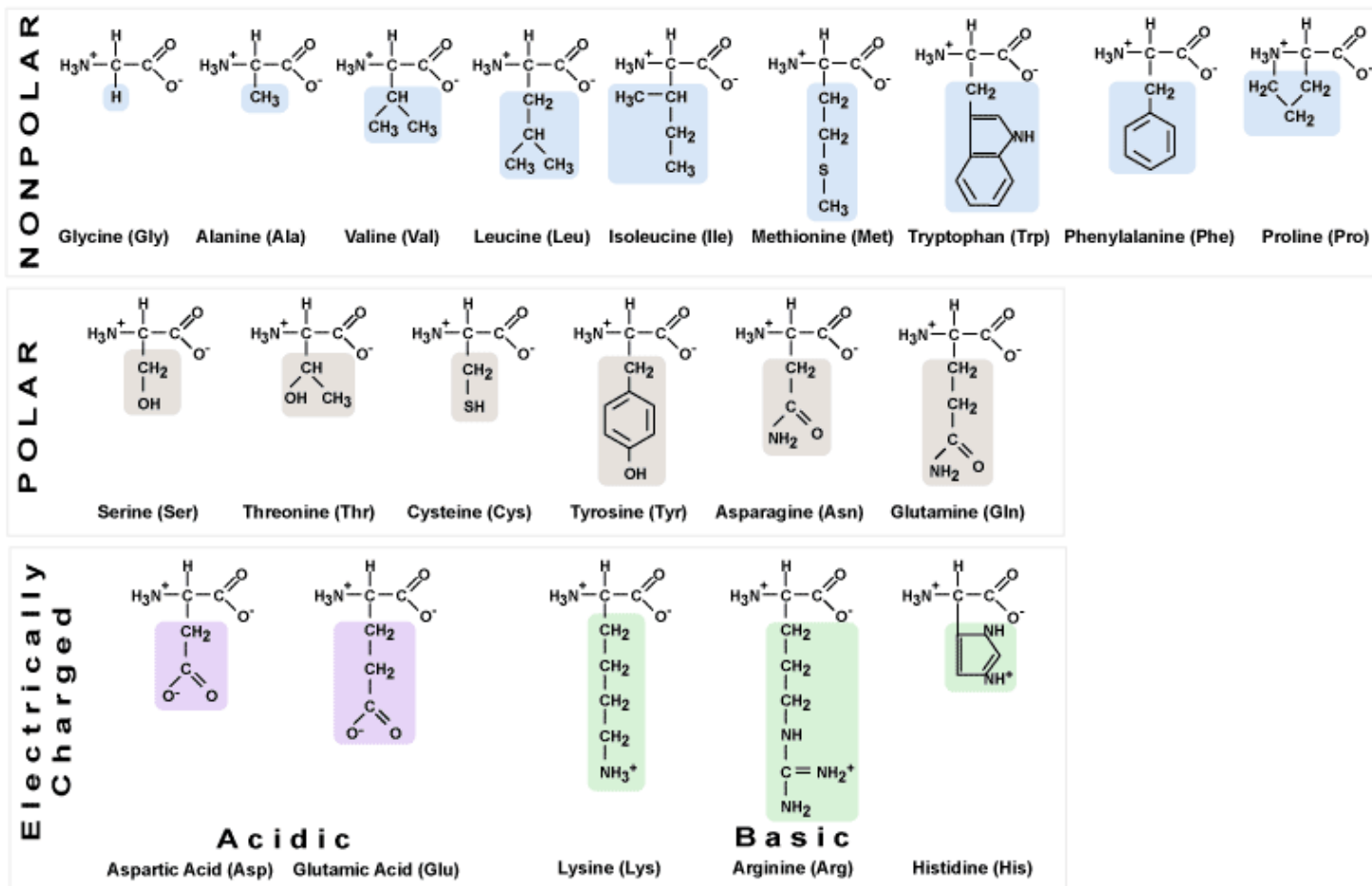


Image (left): <http://www.mcqbiology.com/2012/11/mcq-on-amino-acids.html>

Image (right): <https://www.slideshare.net/MUBOSScz/14-proteins>

20 Typical Monomeric Units



Proteins are Polymer Chains

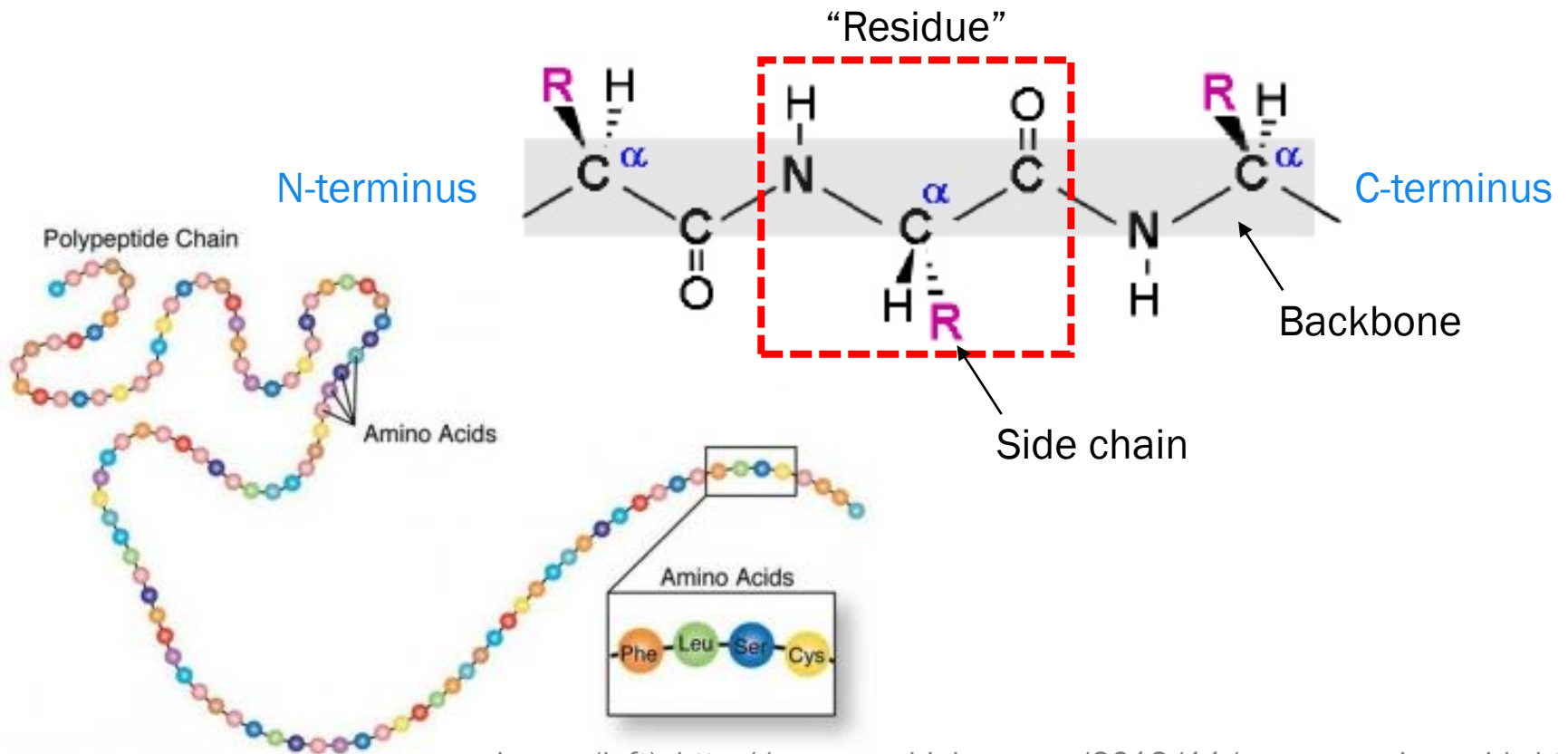
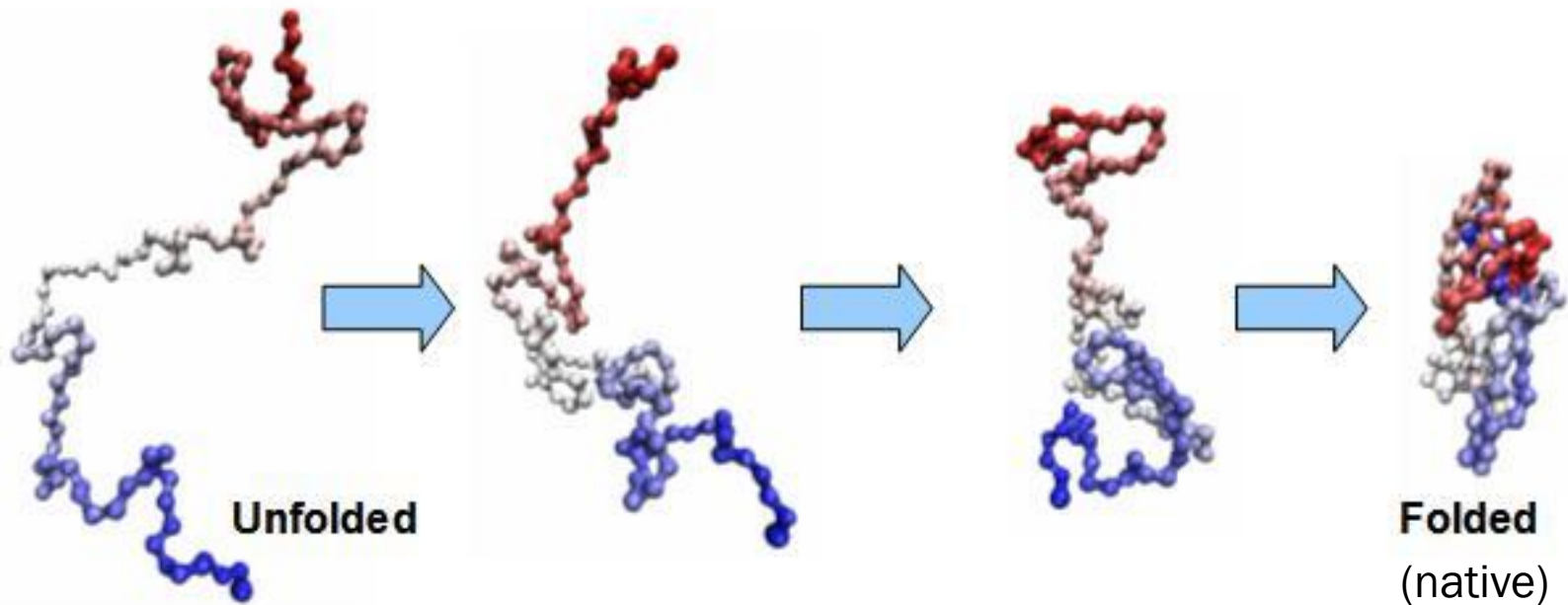


Image (left): <http://www.mcqbiology.com/2012/11/mcq-on-amino-acids.html>

Image (right): <https://www.slideshare.net/MUBOSScz/14-proteins>

Folding: How to Obtain 3D Structure

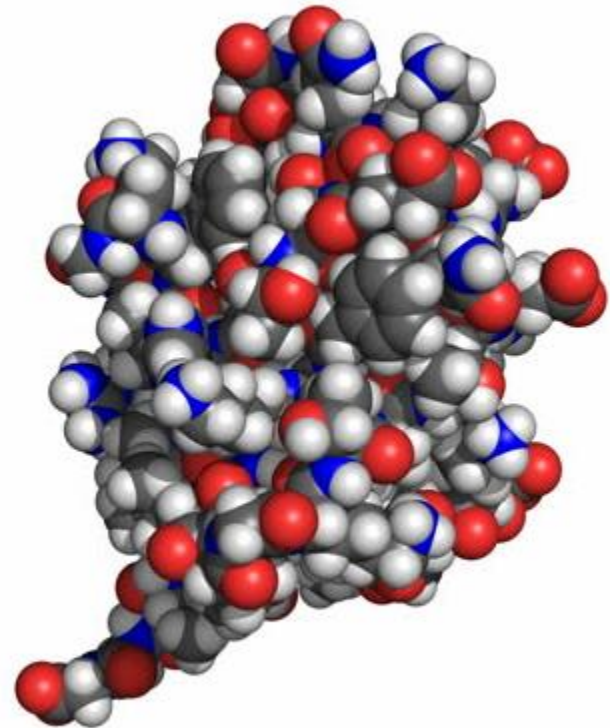


Seeing is believing: <https://youtu.be/yZ2aY5lxEGE>

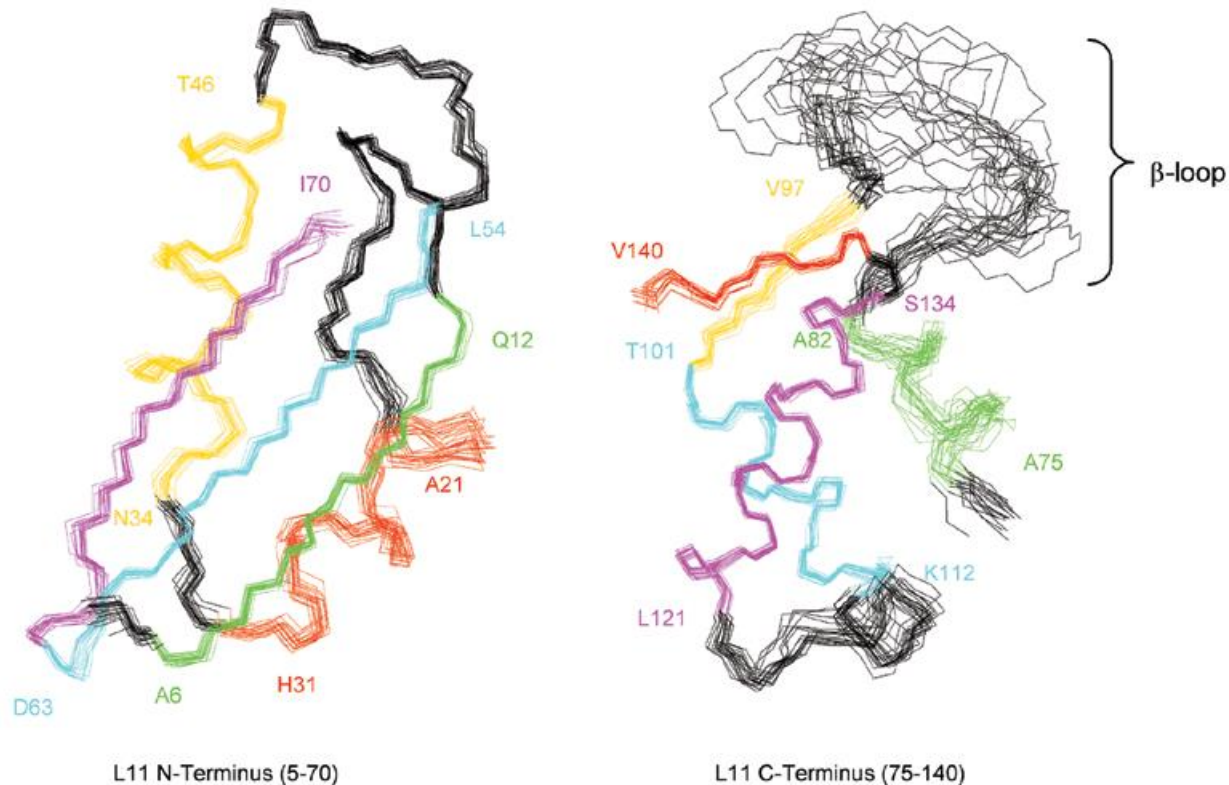
Image: https://inst.eecs.berkeley.edu/~ee127a/book/login/exa_protein_folding.html

Folded Structure is Dense

- Packing density
 - Organic solid: 0.68 – 0.8
 - Proteins: 0.72 – 0.77



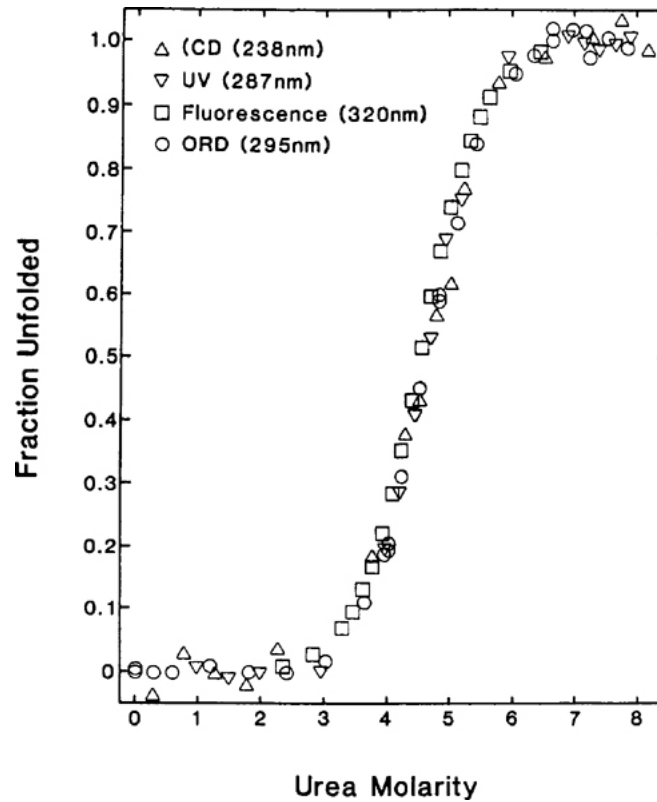
“Frozen” in the Folded State



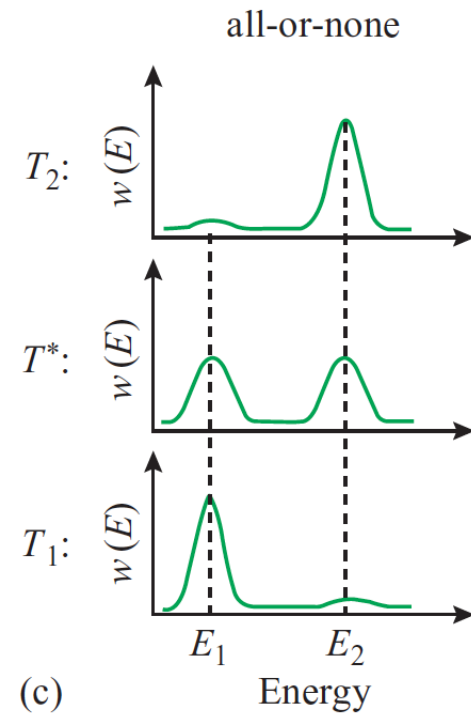
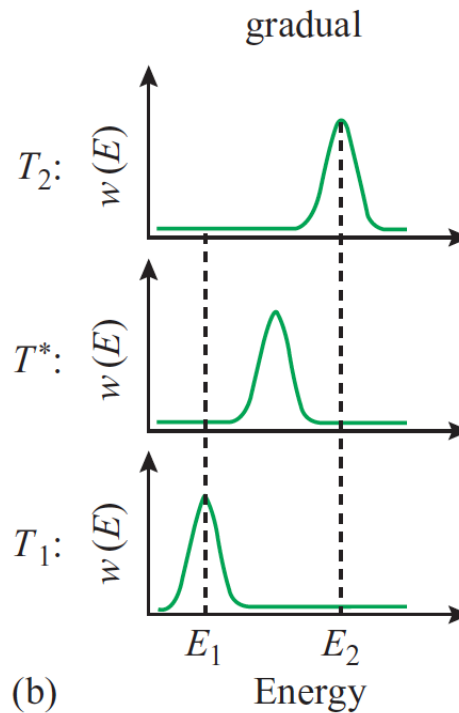
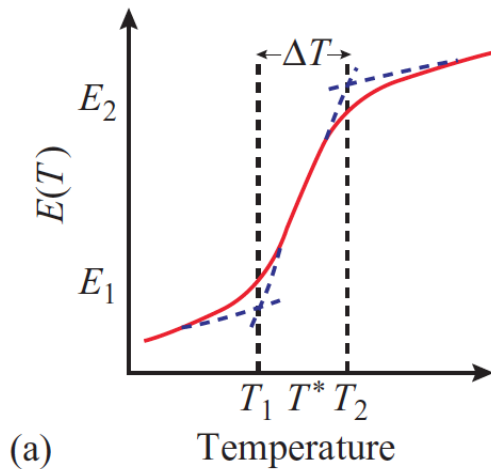
(measurement by NMR)

Image: <http://schwalbe.org.chemie.uni-frankfurt.de/node/684>

Protein Denaturation: S-shaped Curve



First-Order-Like Transition?



van't Hoff Criterion

Assuming two-state folding,

$$P_{\text{fold}} = \frac{e^{-\beta G_{\text{folded}}}}{e^{-\beta G_{\text{folded}}} + e^{-\beta G_{\text{unfolded}}}} = \frac{1}{1 + e^{\beta \Delta G_{\text{fold}}}}$$

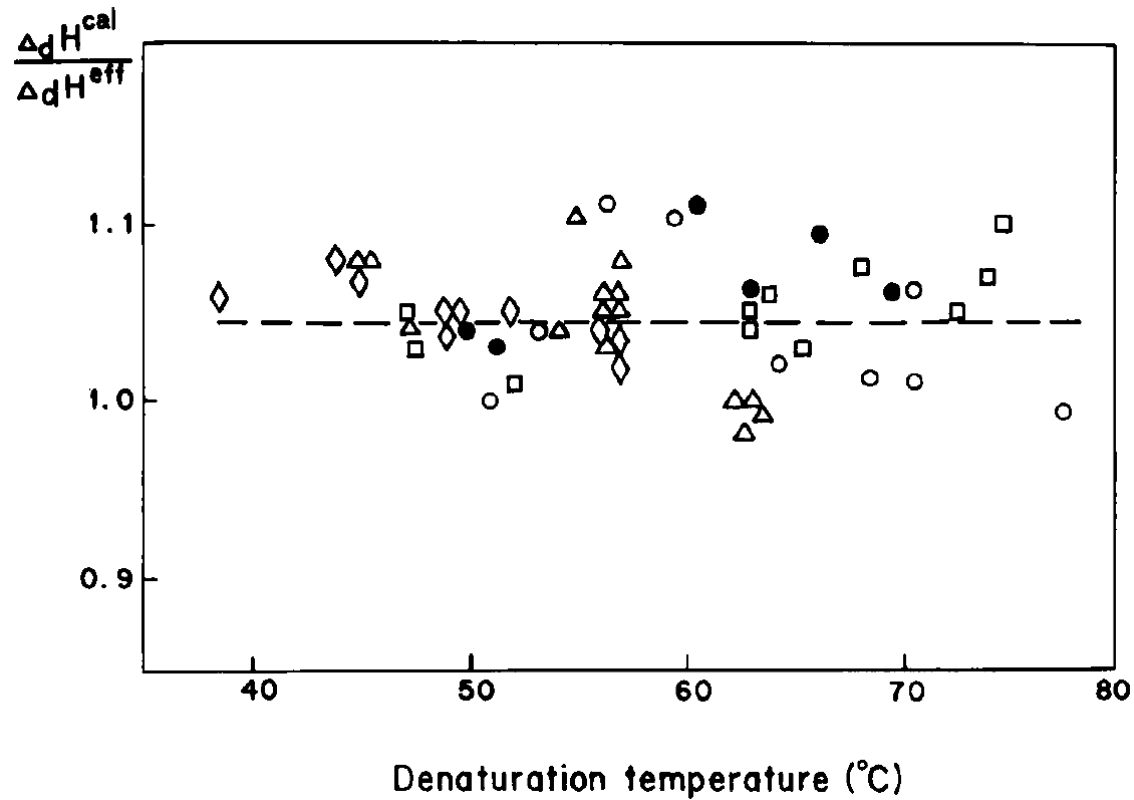
Using the Gibbs-Helmholtz equation,

$$\frac{dP_{\text{fold}}}{dT} = P_{\text{fold}}(1 - P_{\text{fold}}) \frac{\Delta H_{\text{fold}}}{k_B T^2}$$

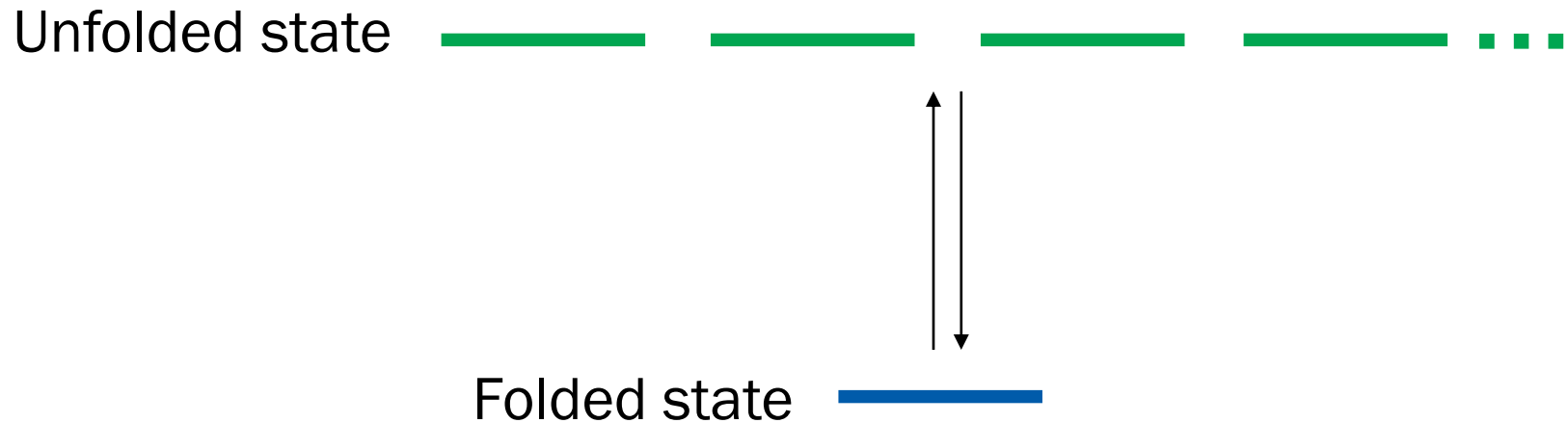
At the transition point ($P_{\text{fold}} = 1/2$),

$$\left. \frac{dP_{\text{fold}}}{dT} \right|_{T=T_f} = \frac{\Delta H_{\text{fold}}}{4k_B T_f^2}$$

Experimentally ...

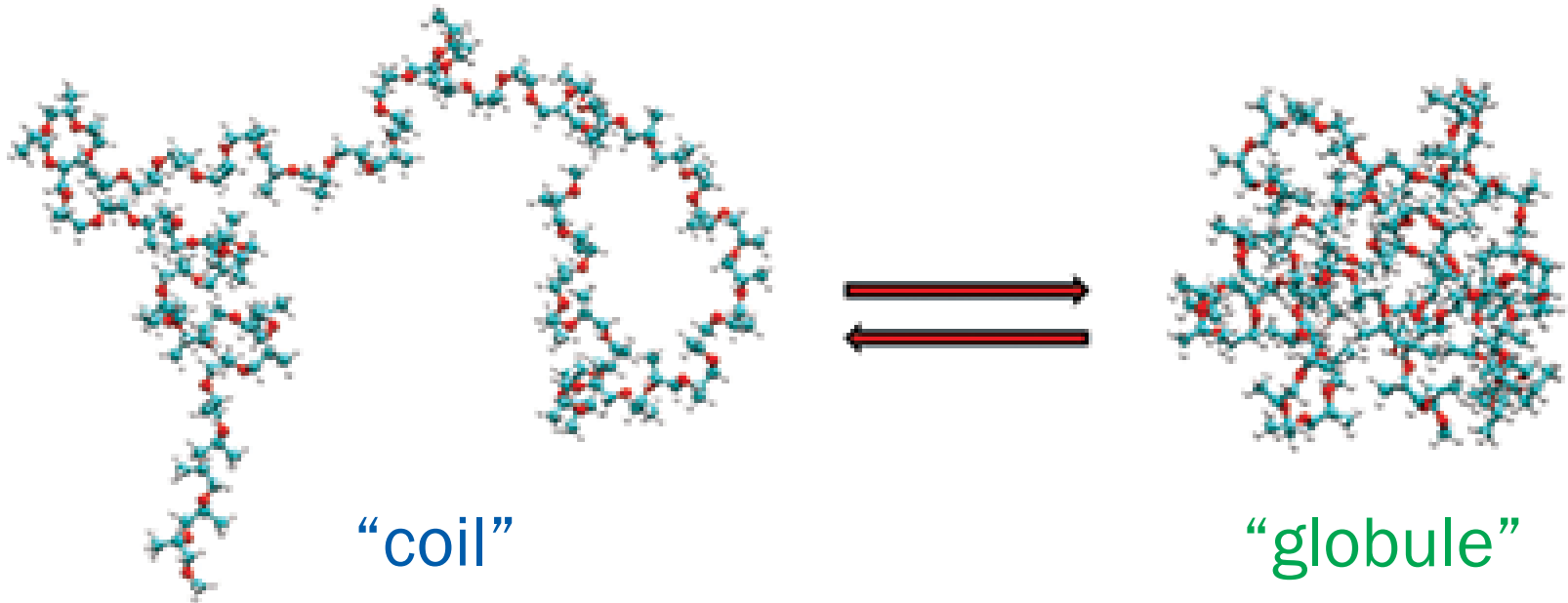


Two-State Model

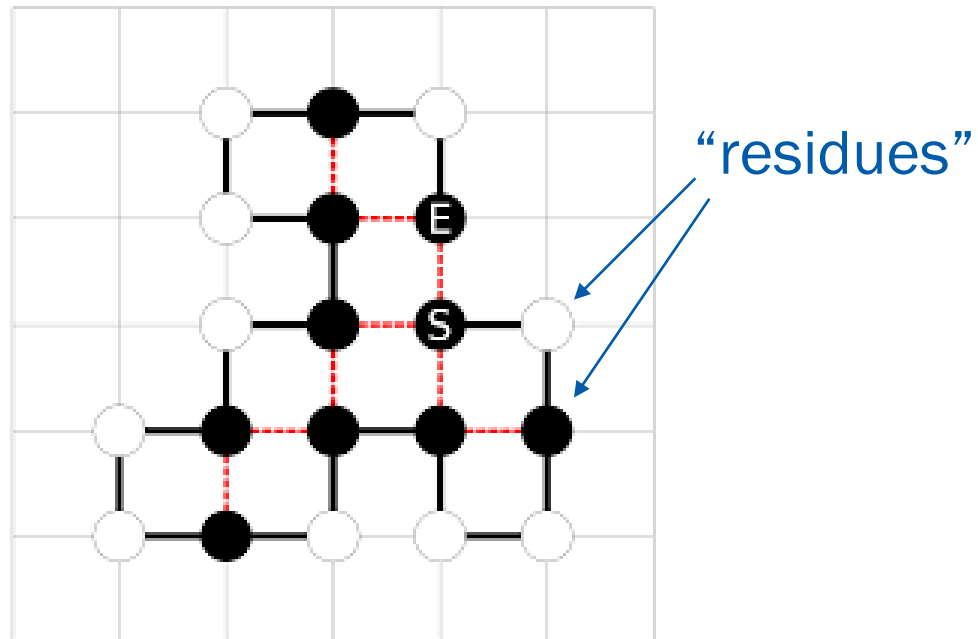


What is the underlying physics?

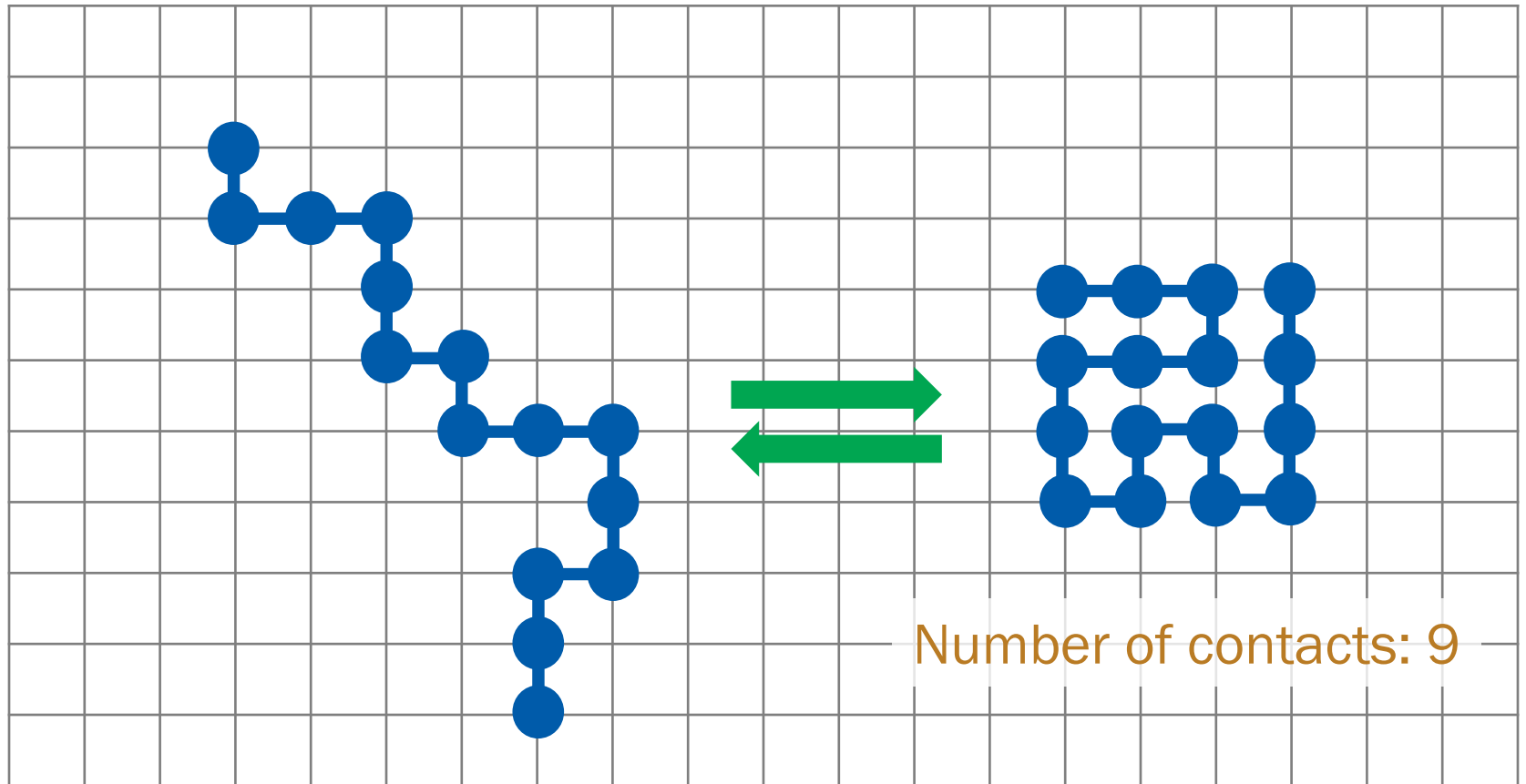
Homopolymers: Coil-Globule Transition



Lattice Model



Coil-Globule Transition

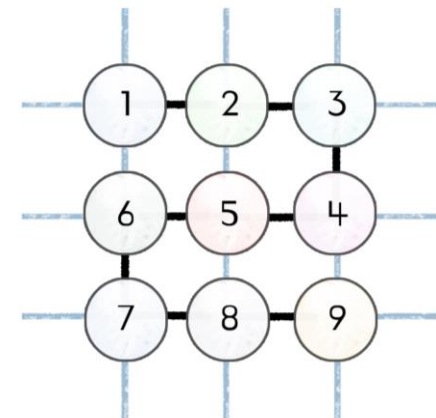
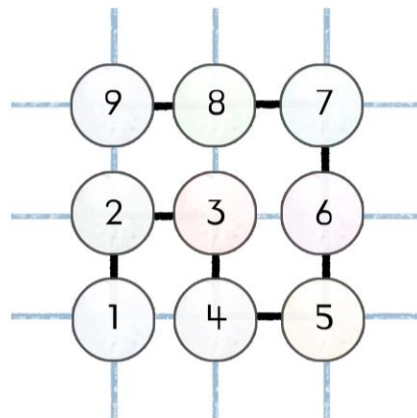
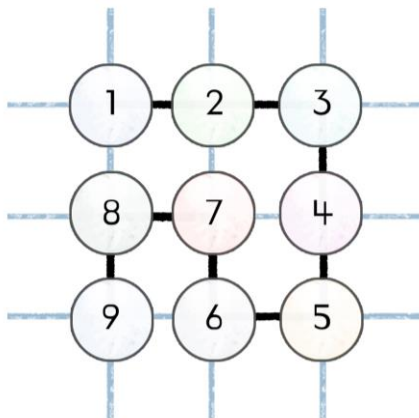
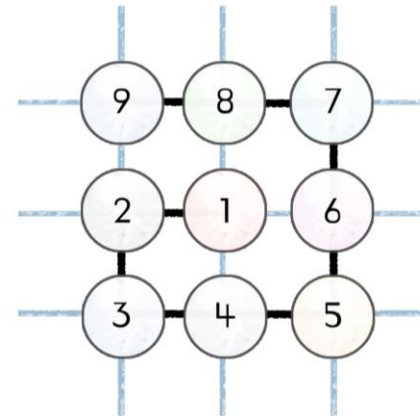
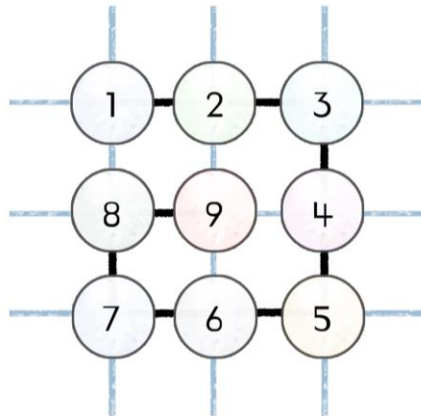


Transition Temperature

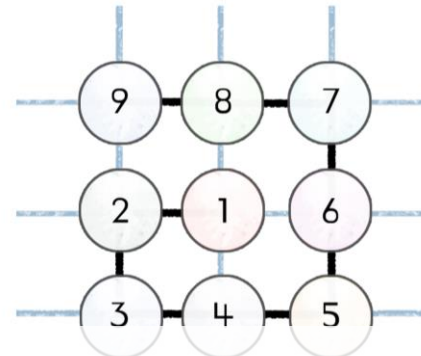
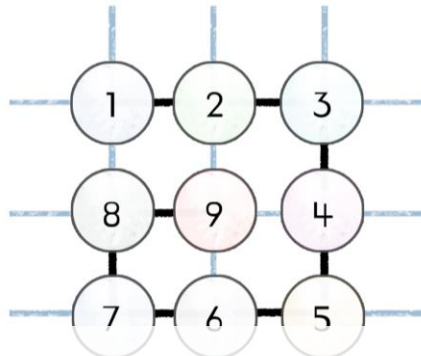
- Coil state: $E_{\text{coil}} = 0$, $S_{\text{coil}} = k_B \ln W_{\text{coil}}$
- Globule state: $E_{\text{globule}} = 9\varepsilon$, $S_{\text{globule}} = 0$ (?)
- Transition temperature T^*

$$T^* = \frac{\Delta E}{\Delta S} = - \frac{9\varepsilon}{k_B \ln W_{\text{coil}}}$$

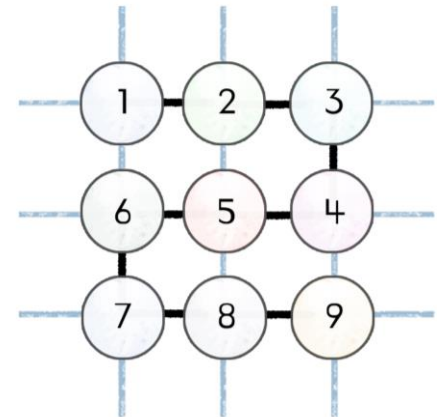
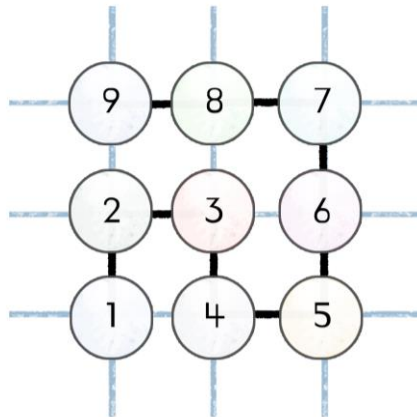
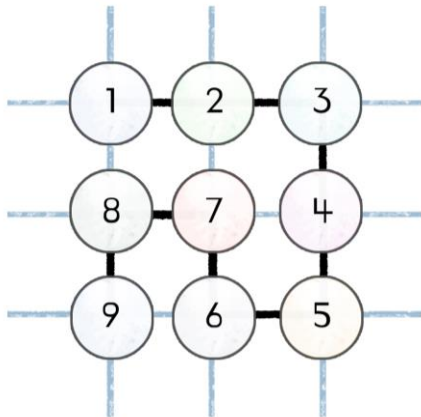
“Globule” Structures of 9-mer



“Globule” Structures of 9-mer

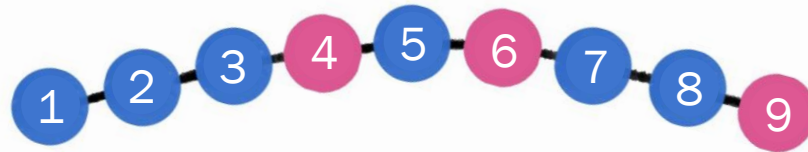


Even the ground state has degeneracy!



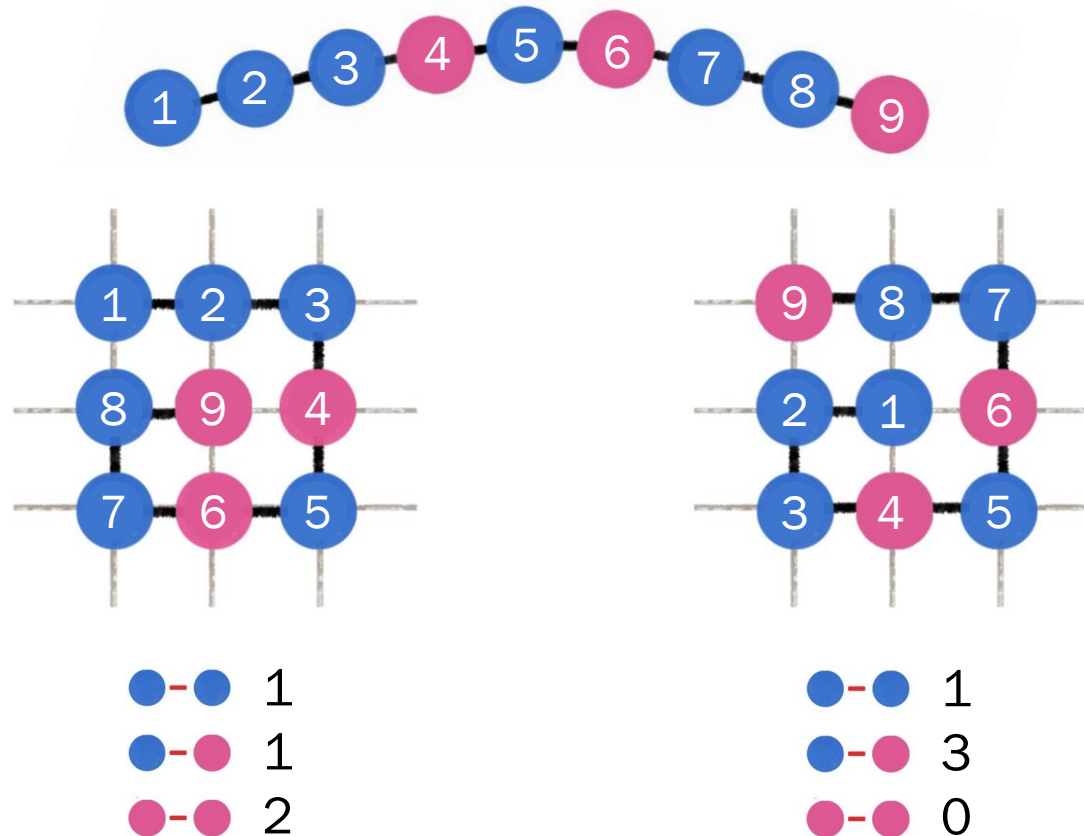
Case of Heteropolymer

Now we have a **sequence**



Number of possible sequences = M^N

Each Structure Gives Different Energy



Energy Distribution

For a sufficiently long heteropolymer, the system energy for a collapsed state is approximately

$$E = \sum_{\substack{i < j \\ \text{in contact}}} \varepsilon_{\sigma(i)\sigma(j)}$$

where $\sigma(i)$ is the unit type of residue i

Random Energy Model

For a random heteropolymer with N units, the collapsed state has an energy distribution of

$$P(E) \propto \exp\left(-\frac{E^2}{J^2 N}\right)$$

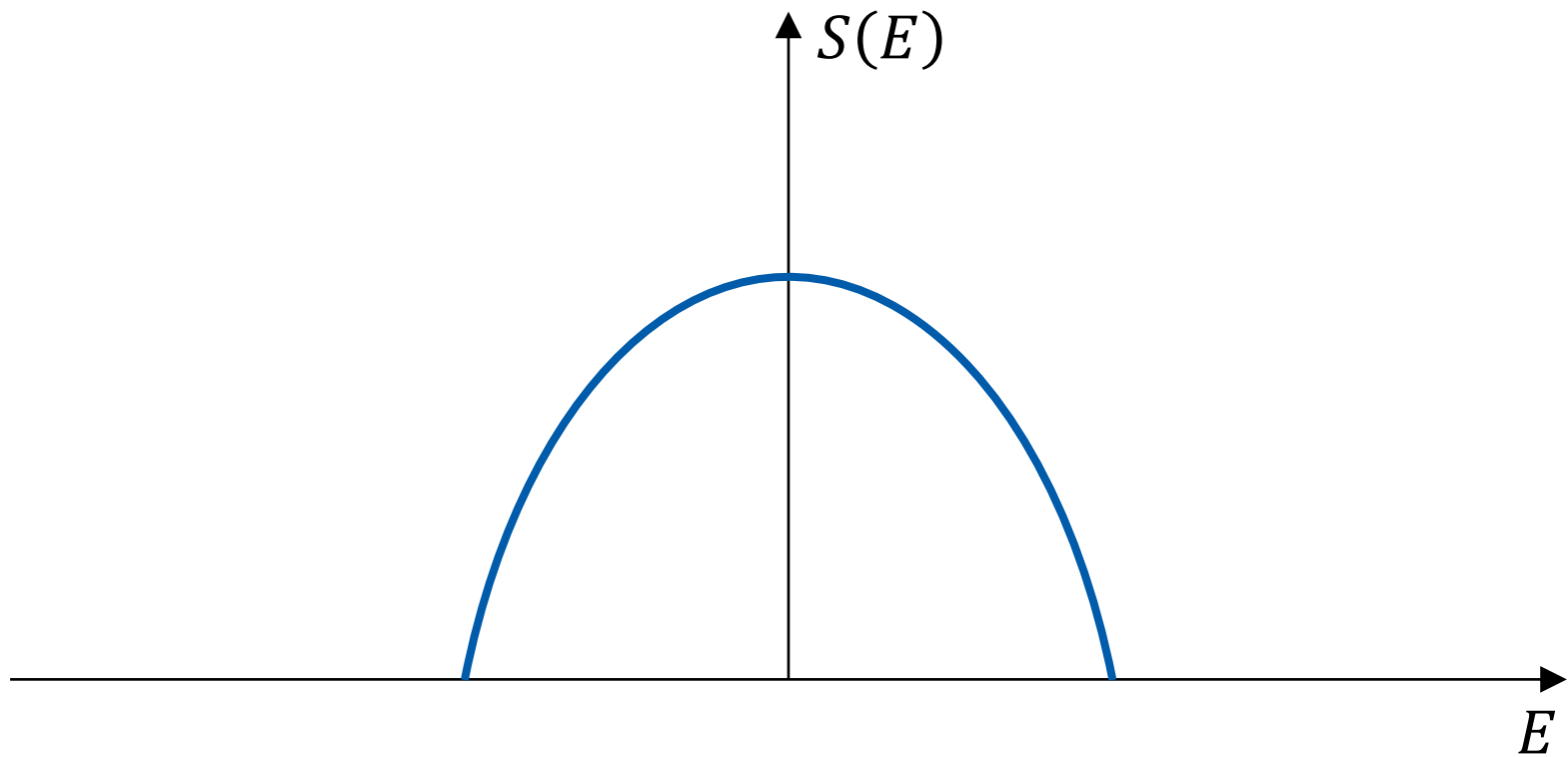
Hence,

$$S(E) = C - \frac{E^2}{J^2 N}$$

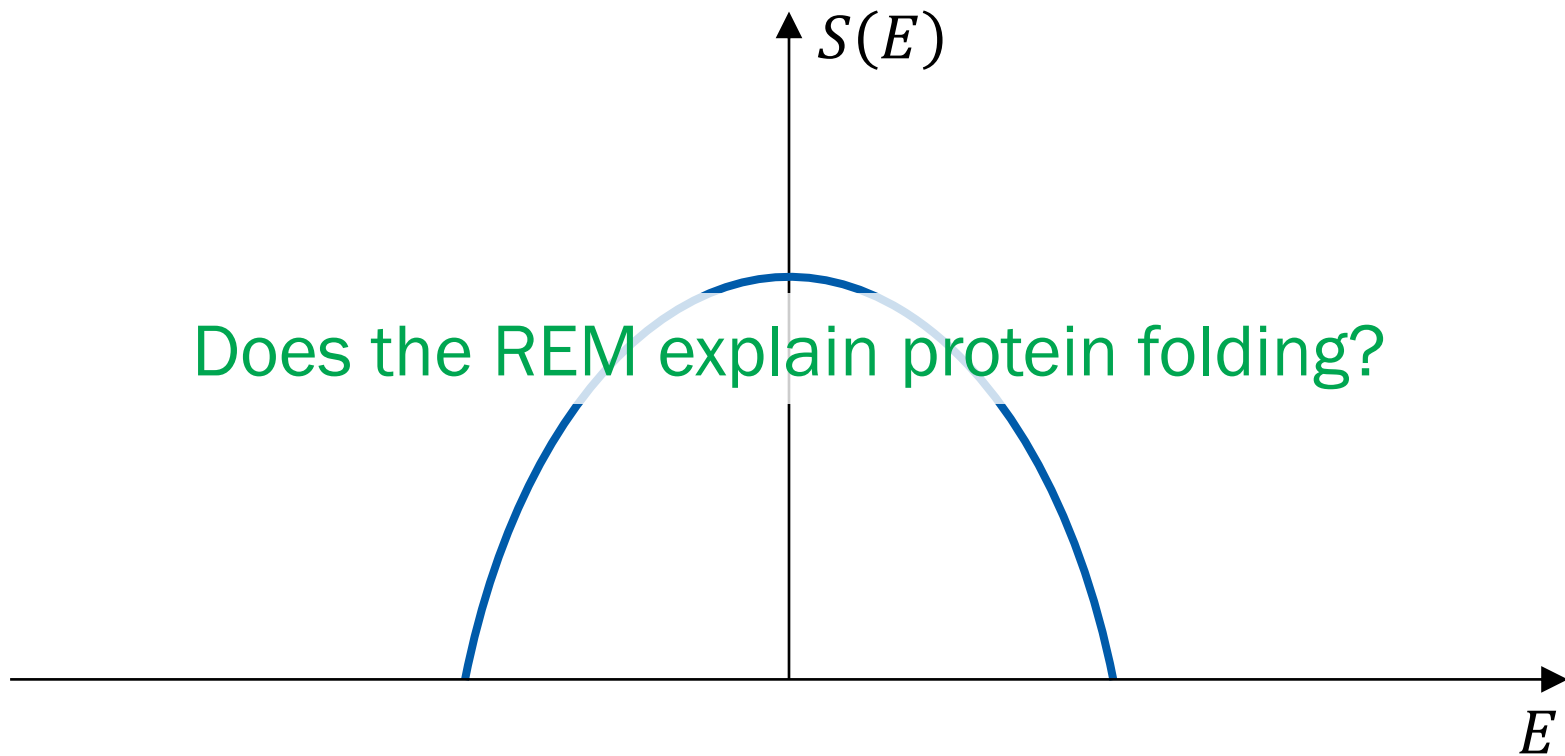


Nobel prize in Physics 2021

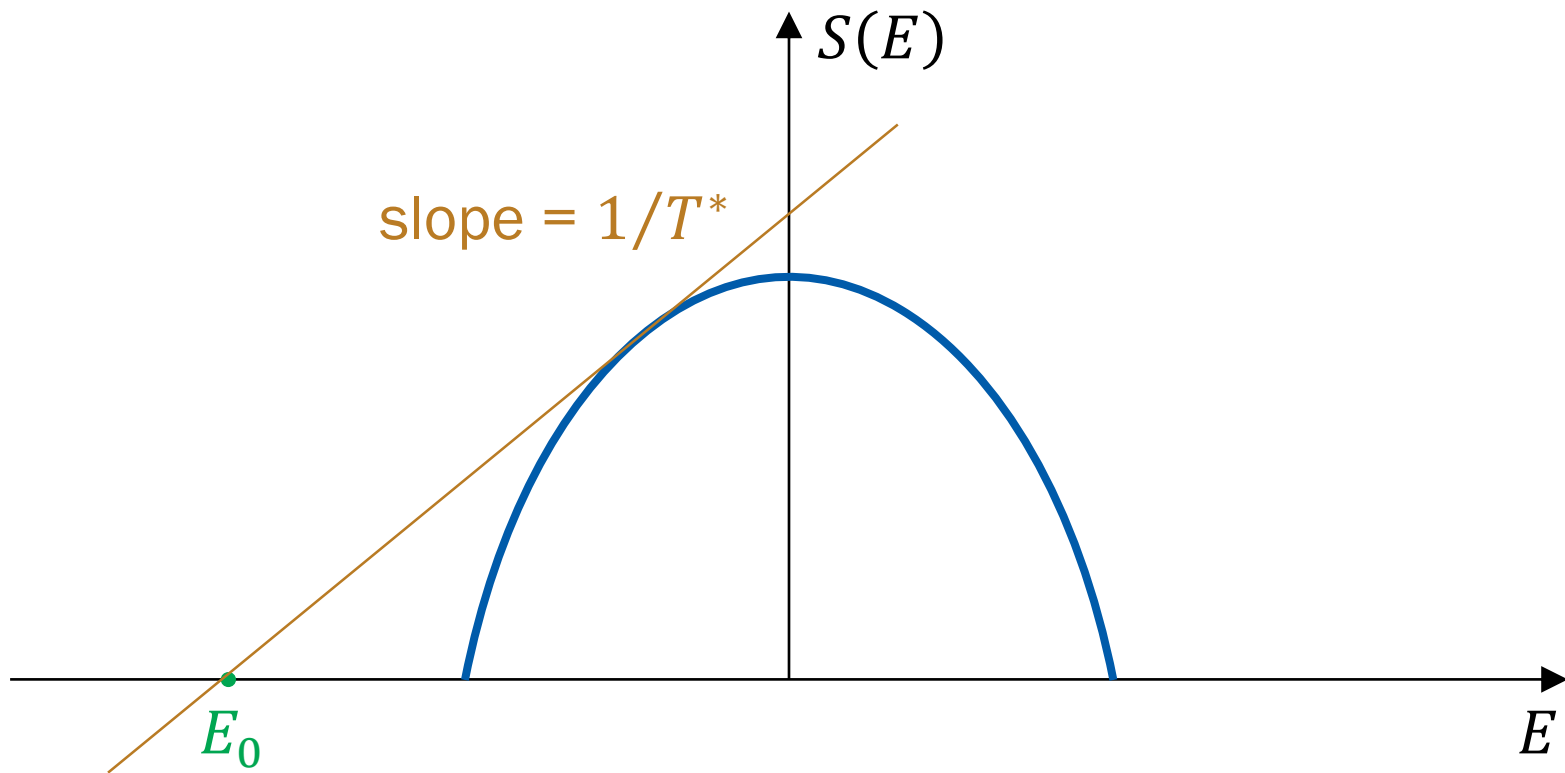
Entropy of REM



Entropy of REM



Final Solution: Energy Gap



Another Representation: Funnel

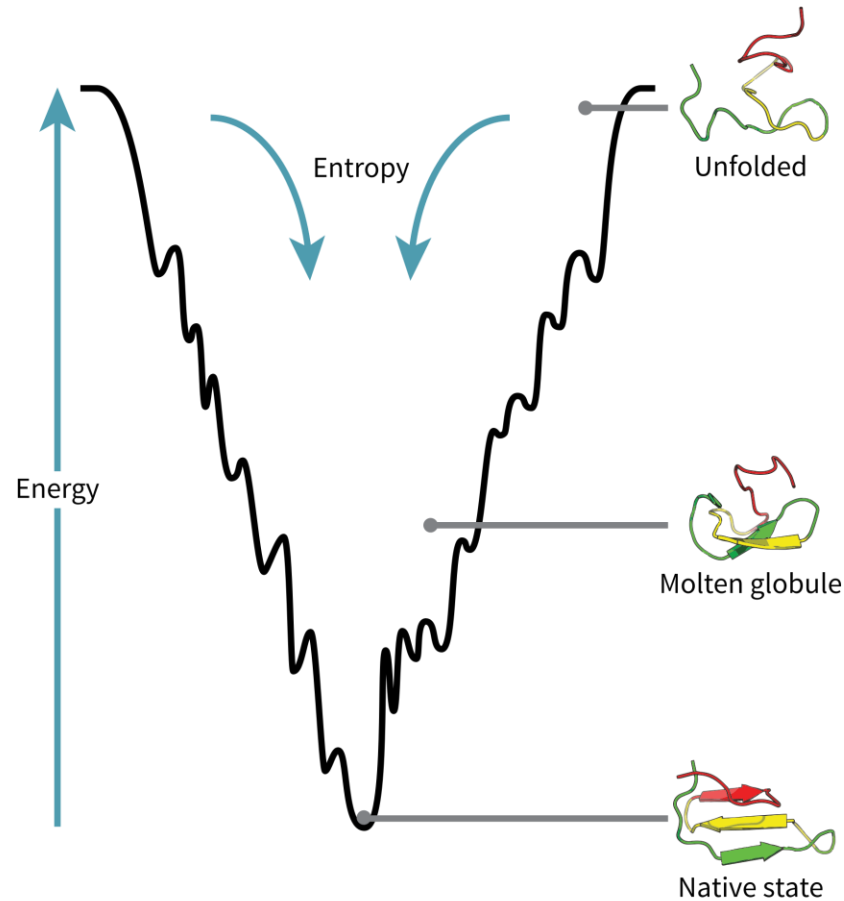
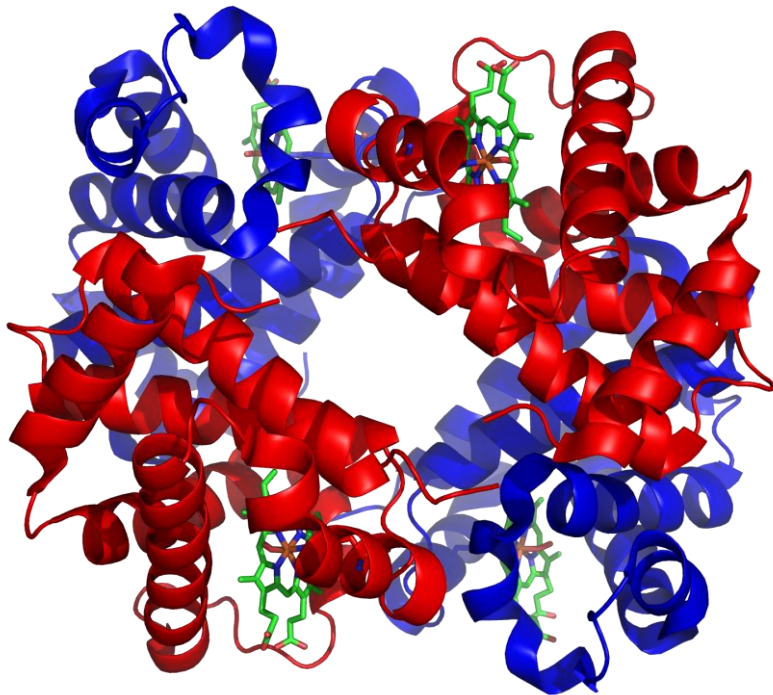


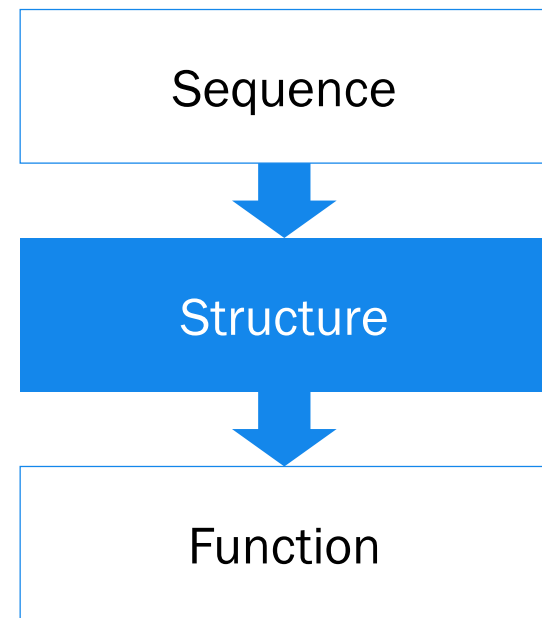
Image: https://en.wikipedia.org/wiki/Protein_folding#/media/File:Folding_funnel_schematic.svg

Why is Folding Important?

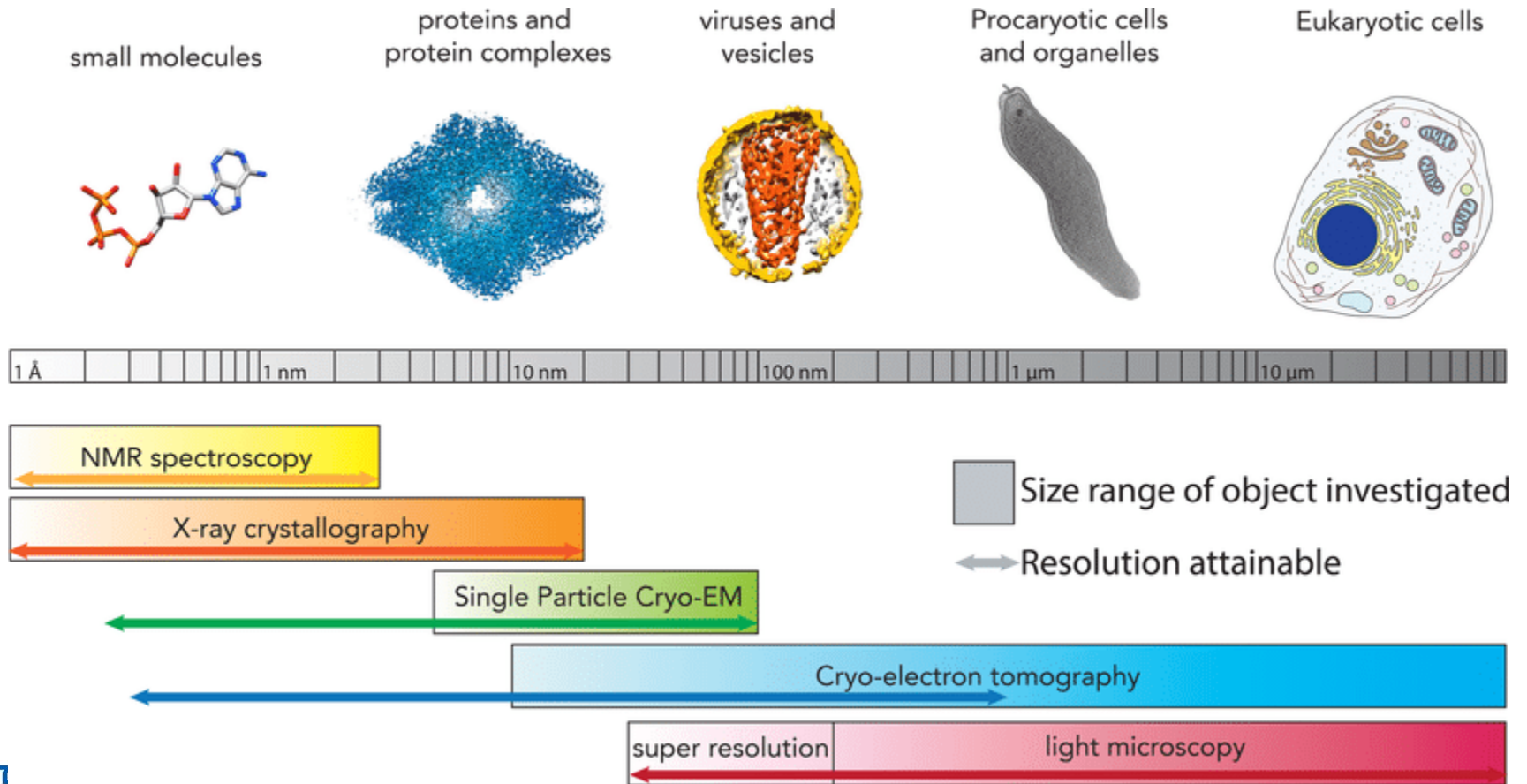


Hemoglobin

Structure-function paradigm



Structural Biology



PDB: Structural Database

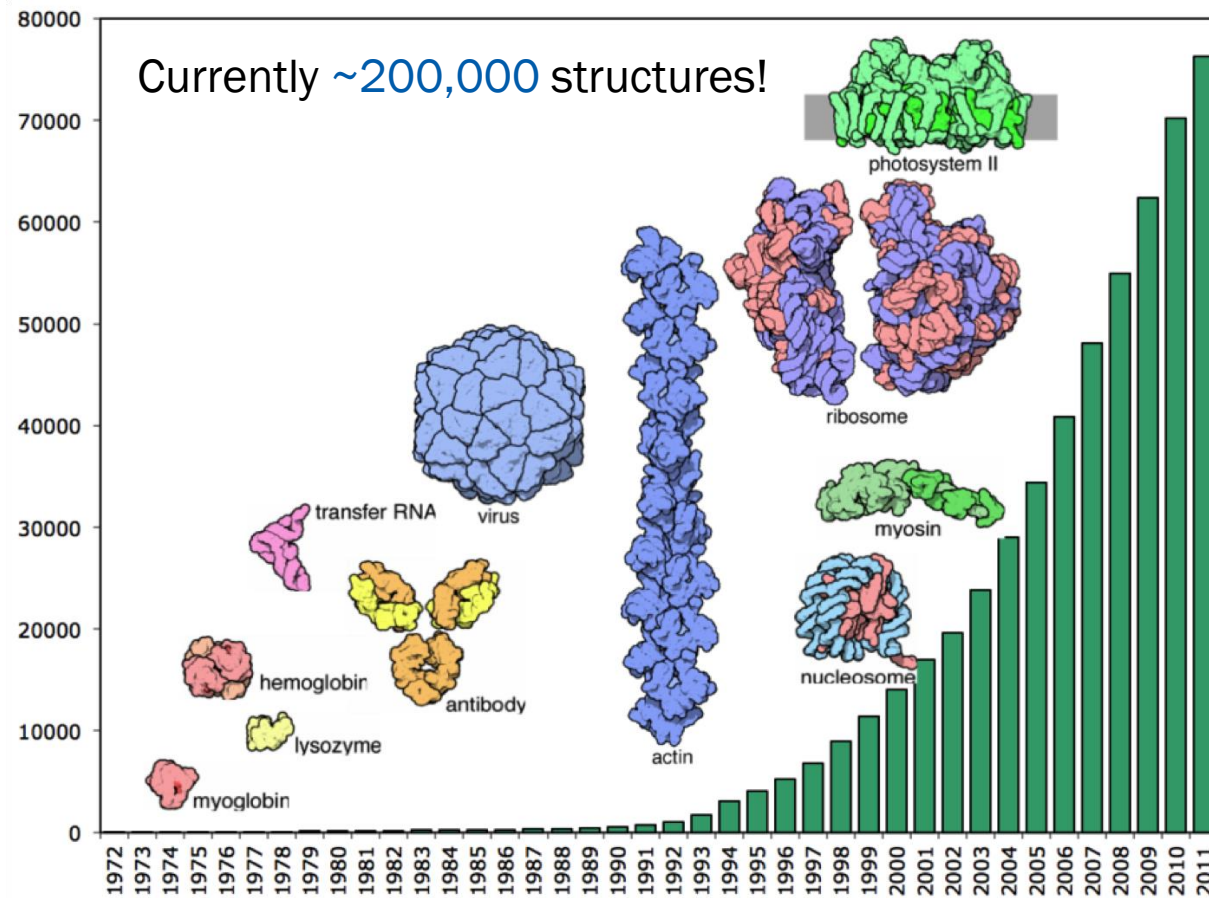
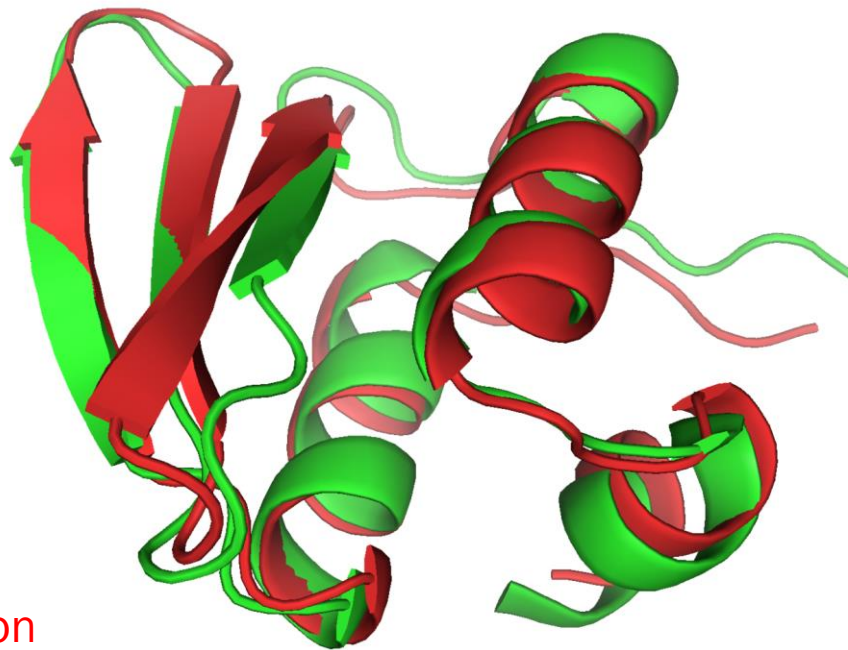


Image: https://cdn.rcsb.org/rcsb-pdb/v2/home/images/panel_search/StatisticsPage.png

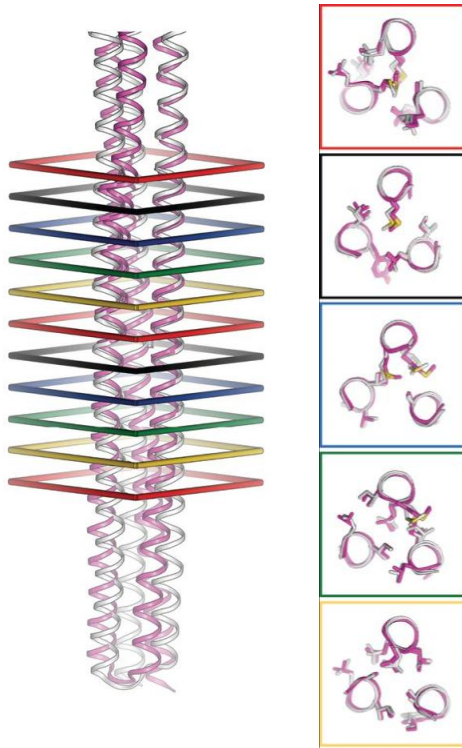
“Computational” Structural Biology



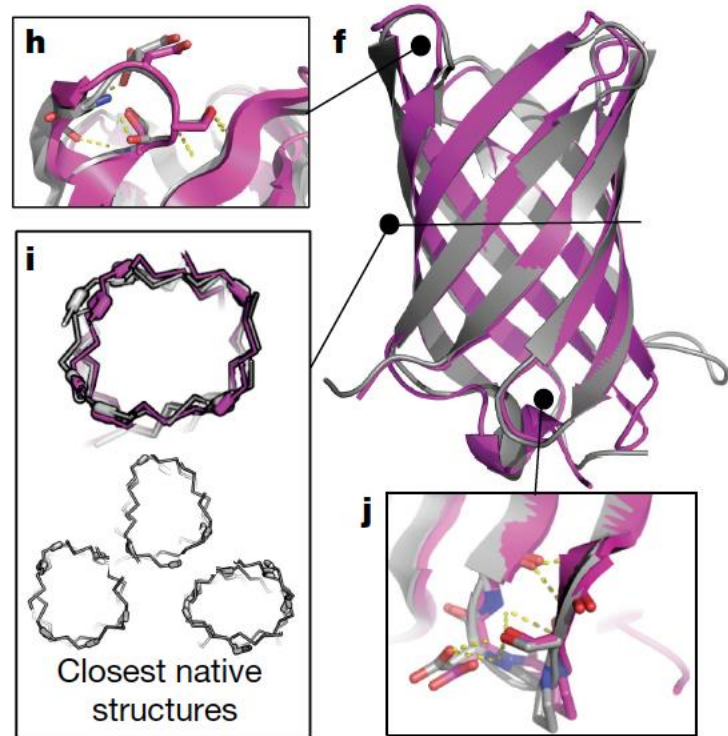
Red: prediction
Green: experiment

De novo Structure Design

α structure



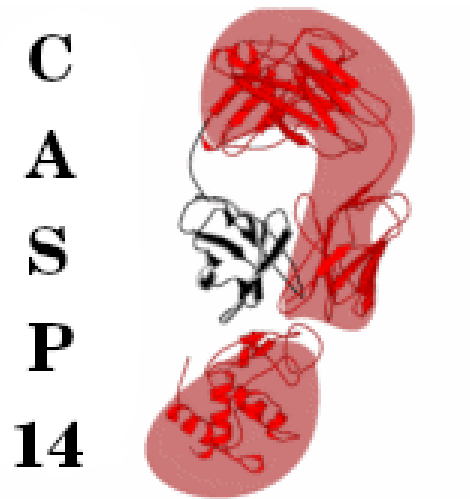
β structure



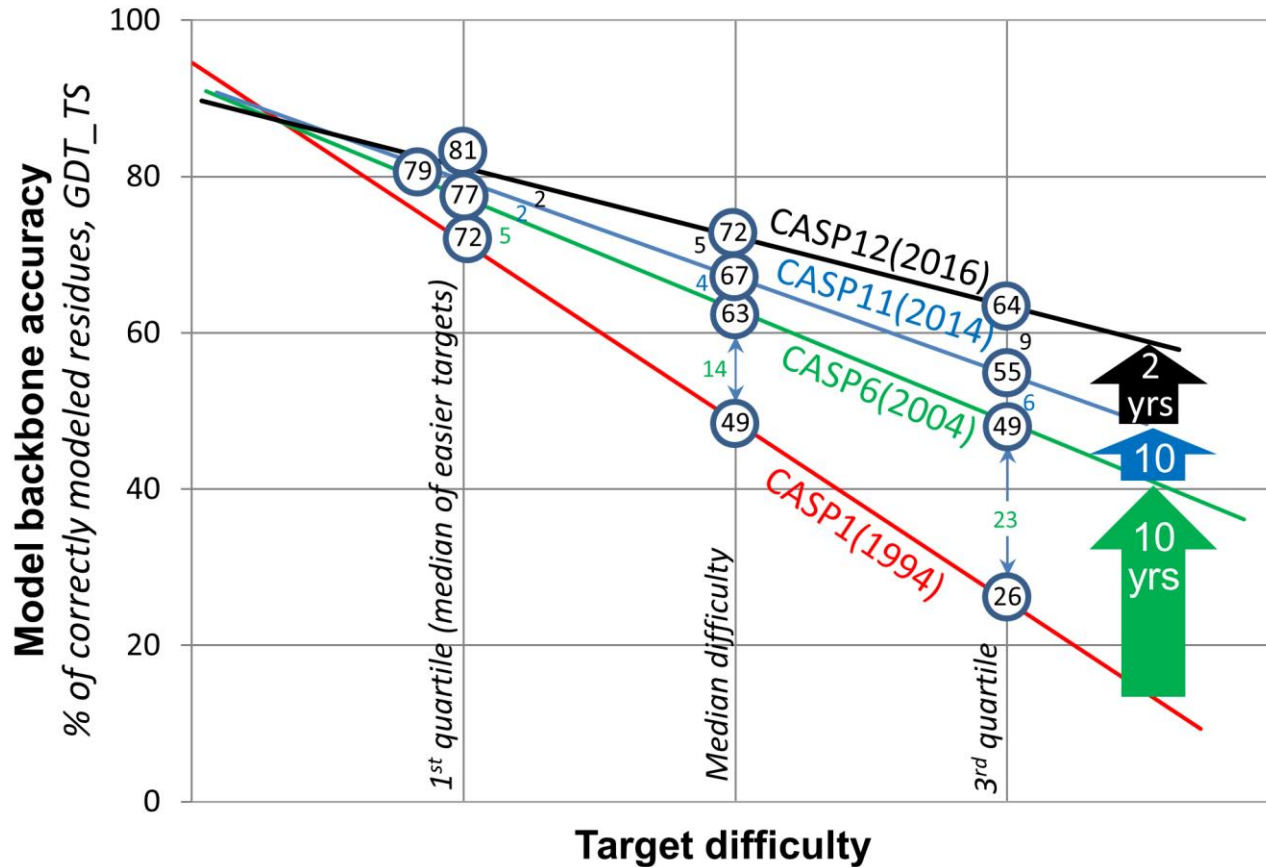
Huang et al., *Science* (2014); Dou et al., *Nature* (2018)

CASP

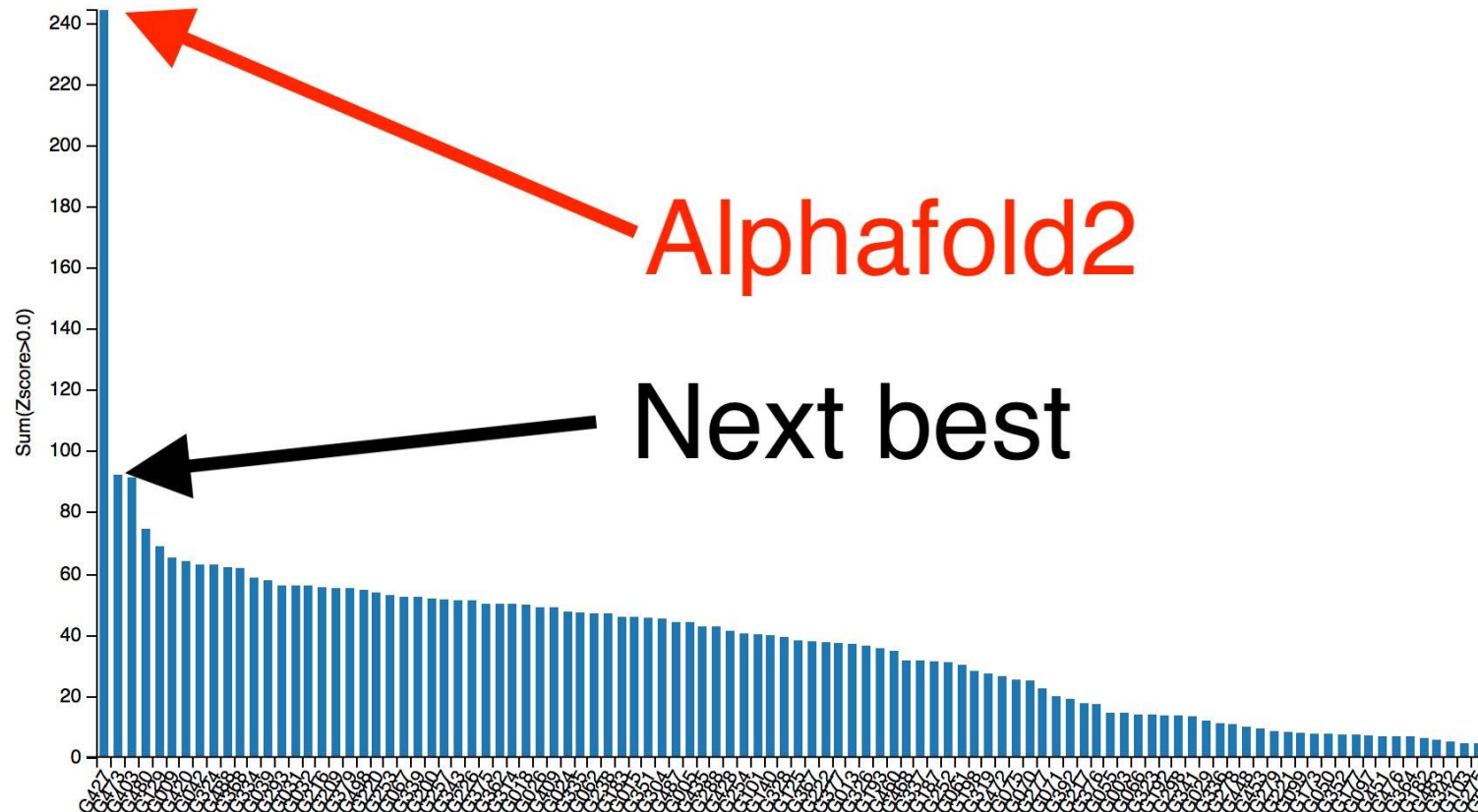
- Critical Assessment of protein Structure Prediction
(<http://predictioncenter.org/>)



Accuracy of Prediction Models



Shocking News from CASP14 (2020)



AlphaFold 2

